Faculty of Management, Social Sciences & Humanities

Department of Management & Finance

General Sir John Kotelawala Defense University

Intake – 42

Semester 3

MTS 210603 - Programming

Group Assignment

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Project: Student Management System for Kotelawala Academy

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1.Abstract

This desktop application is developed for the class teacher at Kotelawala Academy, a tuition center offering Science classes. The system provides a simple and efficient way to record, manage, and view student academic performance for the term.

Key Features:

- Adding new students with name, ID, marks in three subjects (Physics, Chemistry, Biology), and monthly enrollment fee.
- Searching for any student by ID to instantly view total marks, average, and grade (A, B, C, S, or F).
- Automatic grade calculation based on the given scale: A: [75-100], B: [60-75), C: [45-60), S: [35-45), F: [0-35).
- Inventory management and summary of monthly enrollment fees (total fees collected and per-student breakdown).
- Generating a professional "Report Card" as a CSV file for any student, including personal details, subject-wise marks, grades, total/average, and fee summary.
- Displaying a complete list of all students with names and final averages.
- User-friendly Graphical User Interface (GUI) with tabbed navigation.

The system is built using Python with strong emphasis on Object-Oriented Programming (OOP), input validation, error handling, and persistent data storage using JSON files. This project demonstrates fundamental programming concepts, OOP principles, and practical error-handling techniques required for the module.

2.System Design

The system is designed using the Object-Oriented Programming (OOP) approach for better modularity, reusability, and maintainability.

Key Classes:

- Student Class: Represents each student. It stores name, student ID, marks for three subjects, and monthly fee. It includes methods to calculate total marks, average, and assign grade automatically.
- Student Manager Class: Handles all business logic such as adding students, searching, saving/loading data from JSON, calculating total fees, and generating report cards.

Important Code Snippets (with explanation):

Grade calculation logic (inside Student class):

```
def. get grade(self):
 avg = self. average ()
 if 75 <= avg <= 100: return "A"
 elif 60 <= avg < 75: return "B"
# ... (C, S, F)
```

Input validation and error handling (example from add_student method):

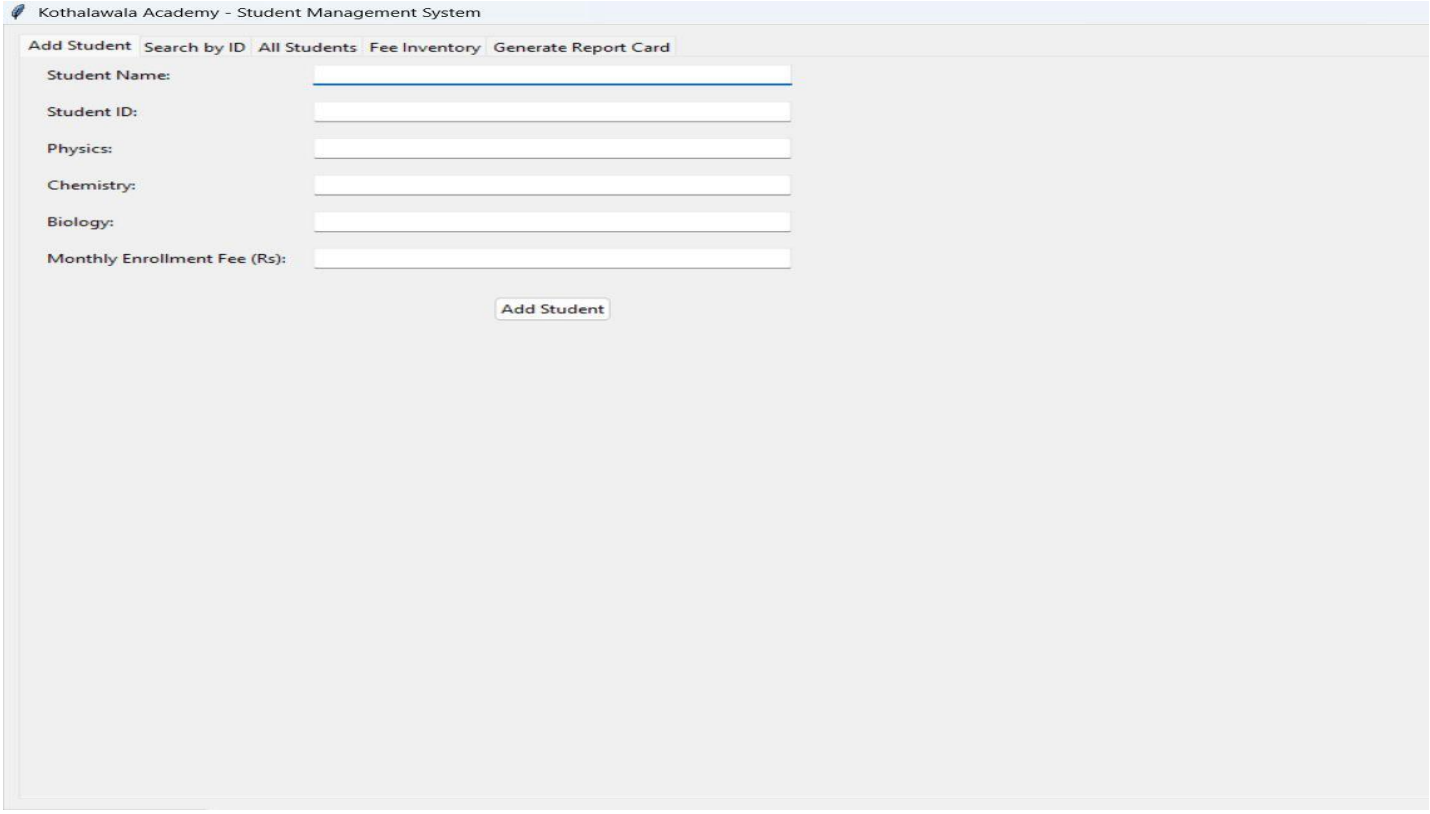
```
if any (s. student_id == student_id for s in self. students):
 raise ValueError ("Student ID already exists!")
if not (0 <= float(marks) <= 100):
 raise ValueError ("Marks must be between 0 and 100!")
```

Special Validations Implemented:

- Required fields (name and ID cannot be empty).
- Marks must be numeric and between 0 and 100.
- Student ID must be unique (duplicate check).
- Monthly fee cannot be negative.

- Add Student
- Search by ID
- All Students
- Fee Inventory
- Generate Report Card

Main Window



Adding a New Student

Open the “Add Student” tab → Fill in Name, Student ID, marks for Physics/Chemistry/Biology, and Monthly Fee → Click “Add Student”.

Success message appears and data is saved automatically.

Searching for a Student

Go to “Search by ID” tab → Enter Student ID → Click “Search”.

Complete details (marks, total, average, grade, fee) are displayed instantly.

Viewing All Students

Open “All Students” tab → Click “Refresh List” to see every student with name, ID, average, and grade.

Fee Inventory Management

Go to “Fee Inventory” tab → Click “Refresh Fee Summary”.

Shows total monthly fees collected and individual student fee breakdown.

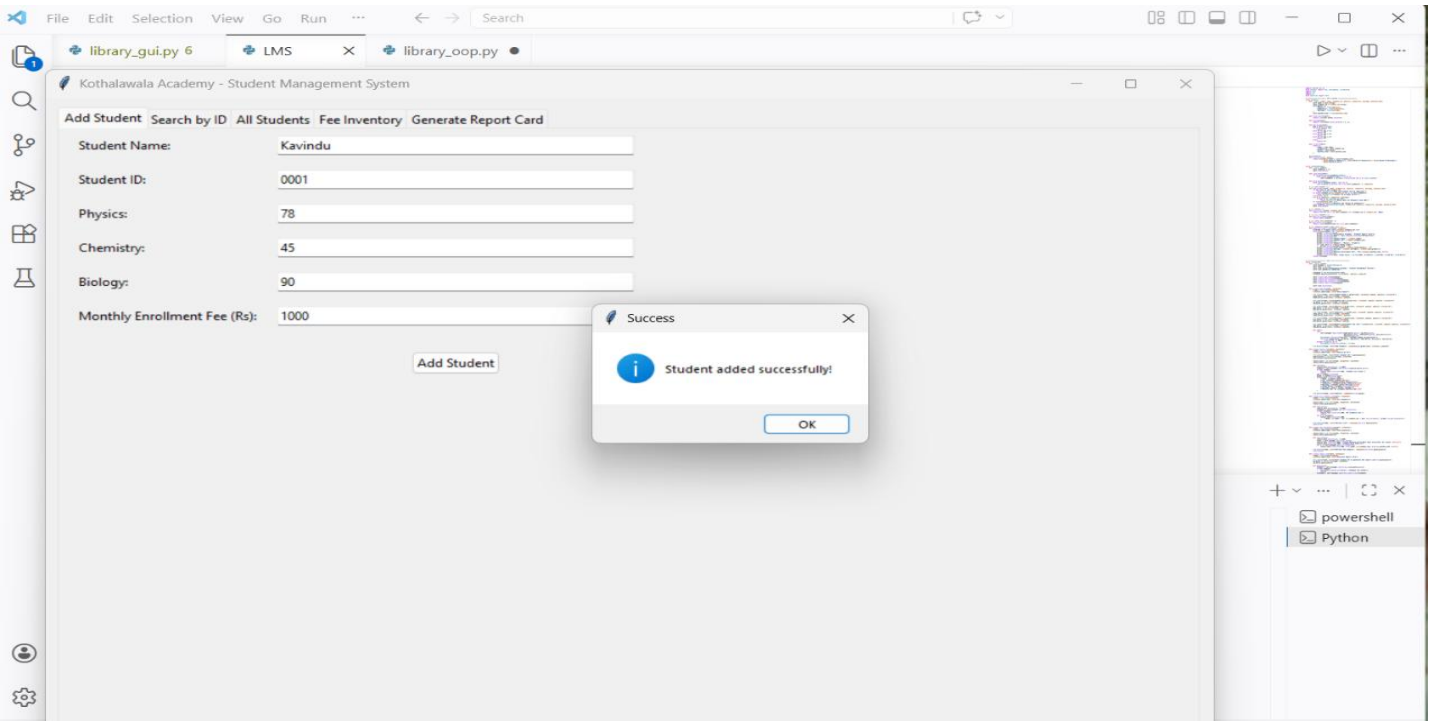
Generating Report Card

Go to “Generate Report Card” tab → Enter Student ID → Click “Generate CSV Report Card”.

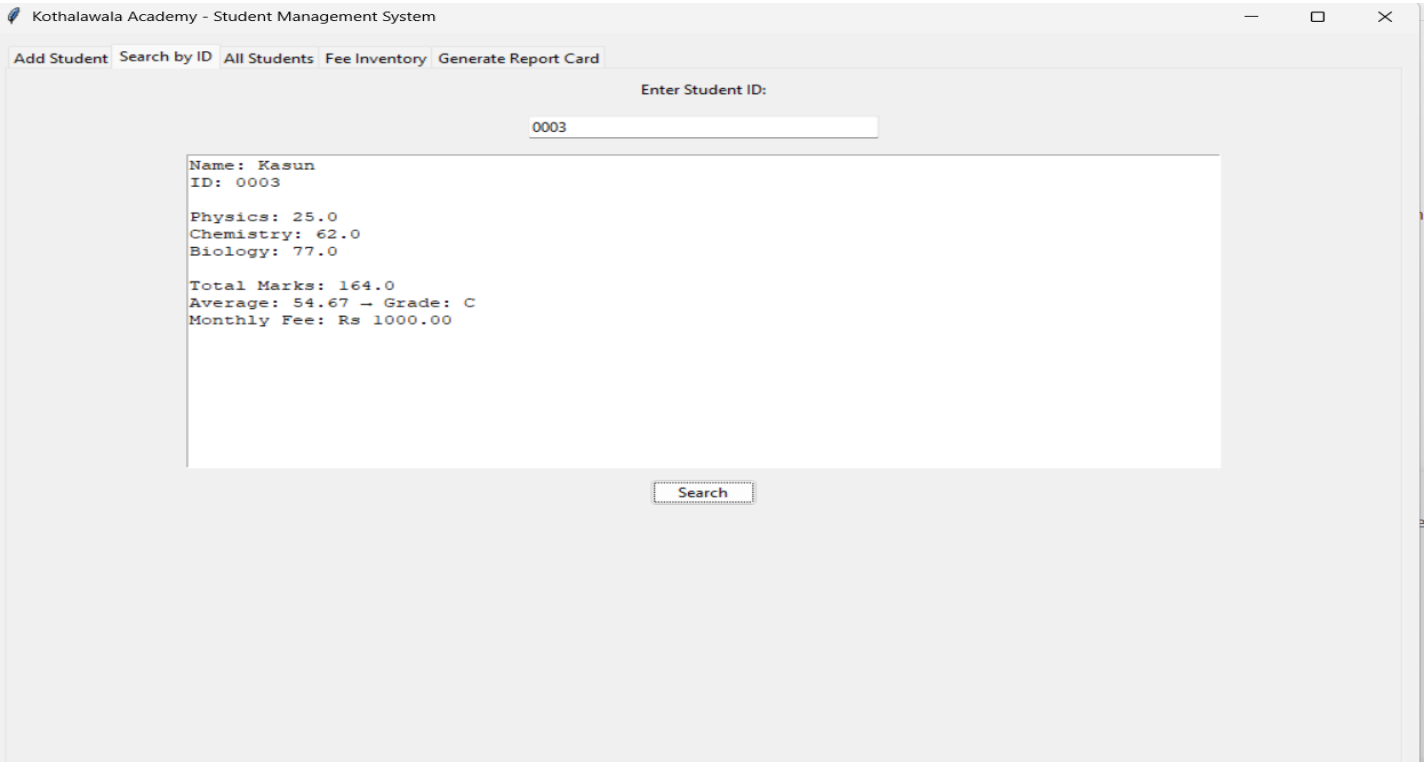
A CSV file (e.g., report_card_ABC123.csv) is created automatically. It can be opened with Excel.Example Usage Scenario:

A teacher adds 5 students, searches for one student to check performance, views the fee summary, and generates report cards for parents.

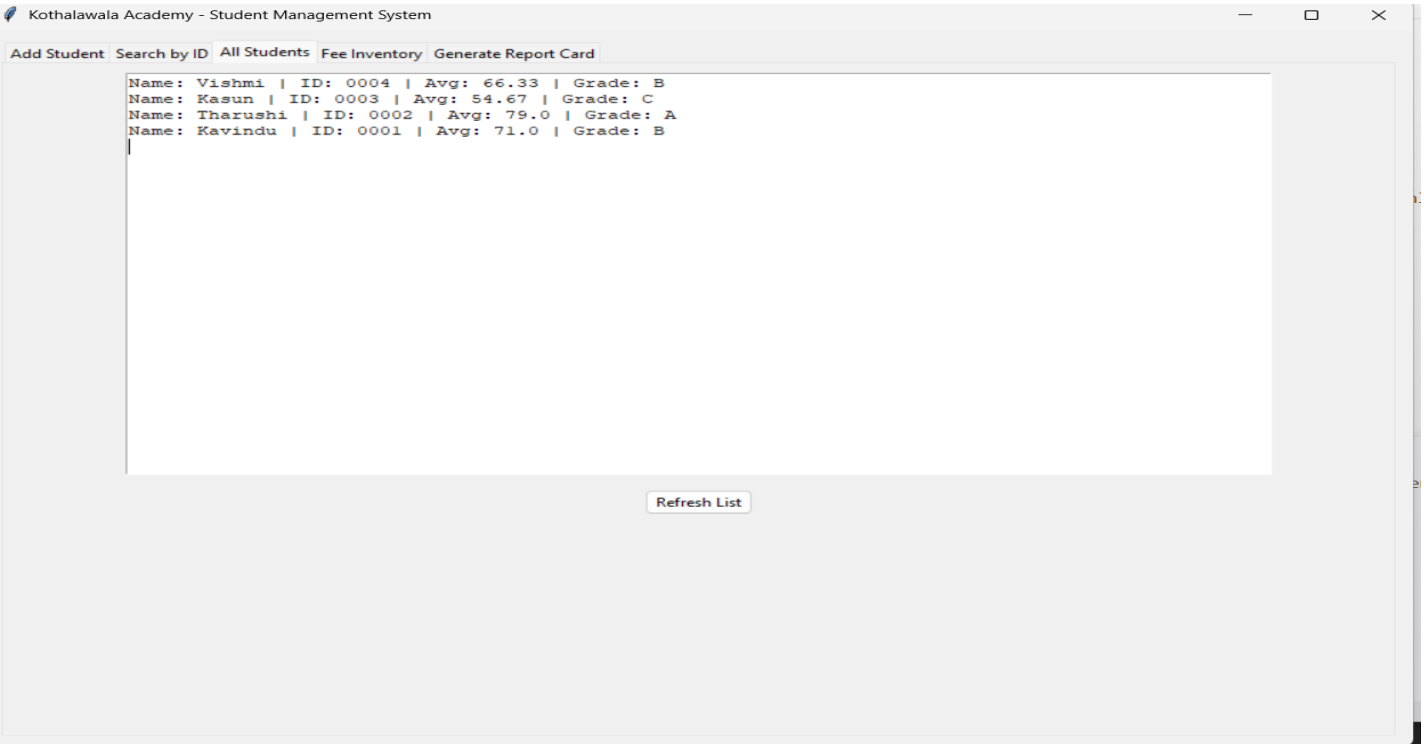
Add Student



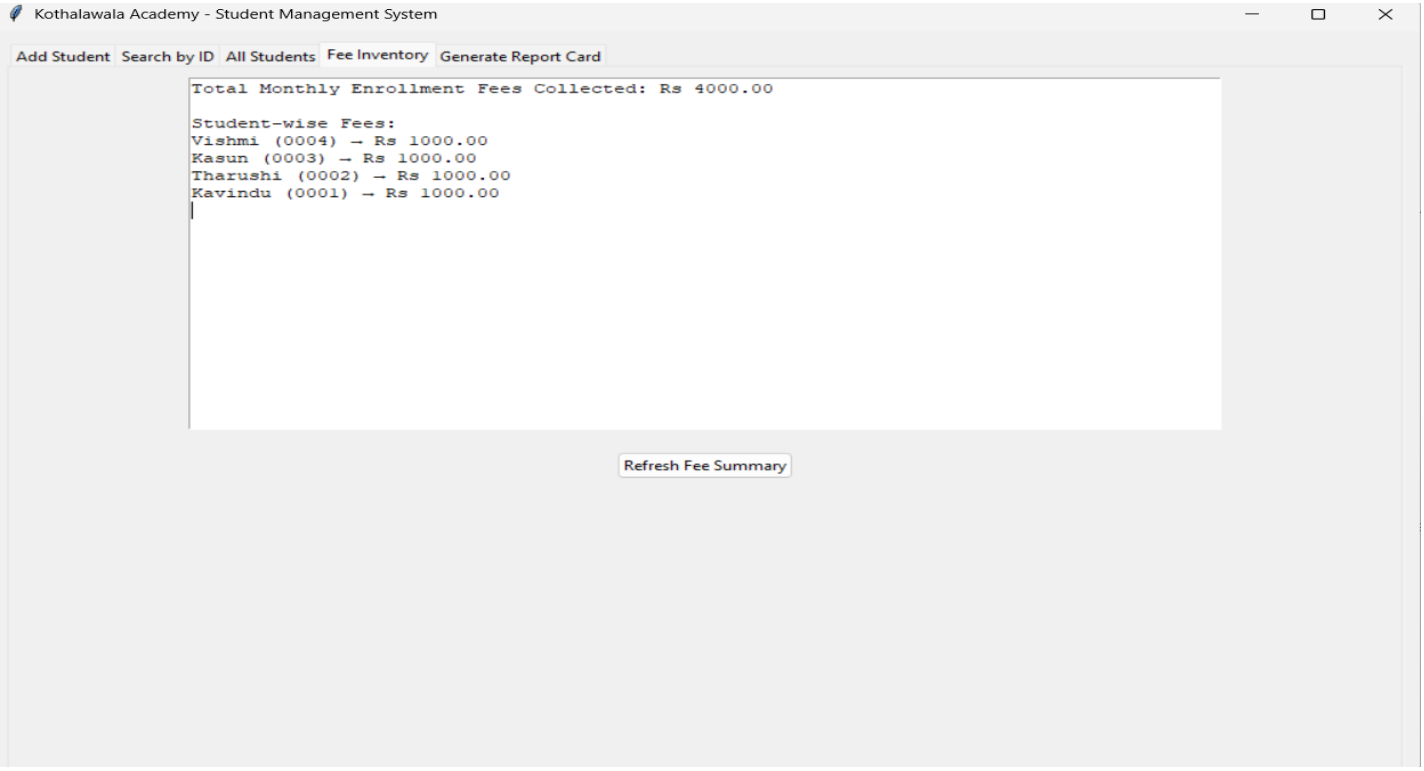
Search Tab



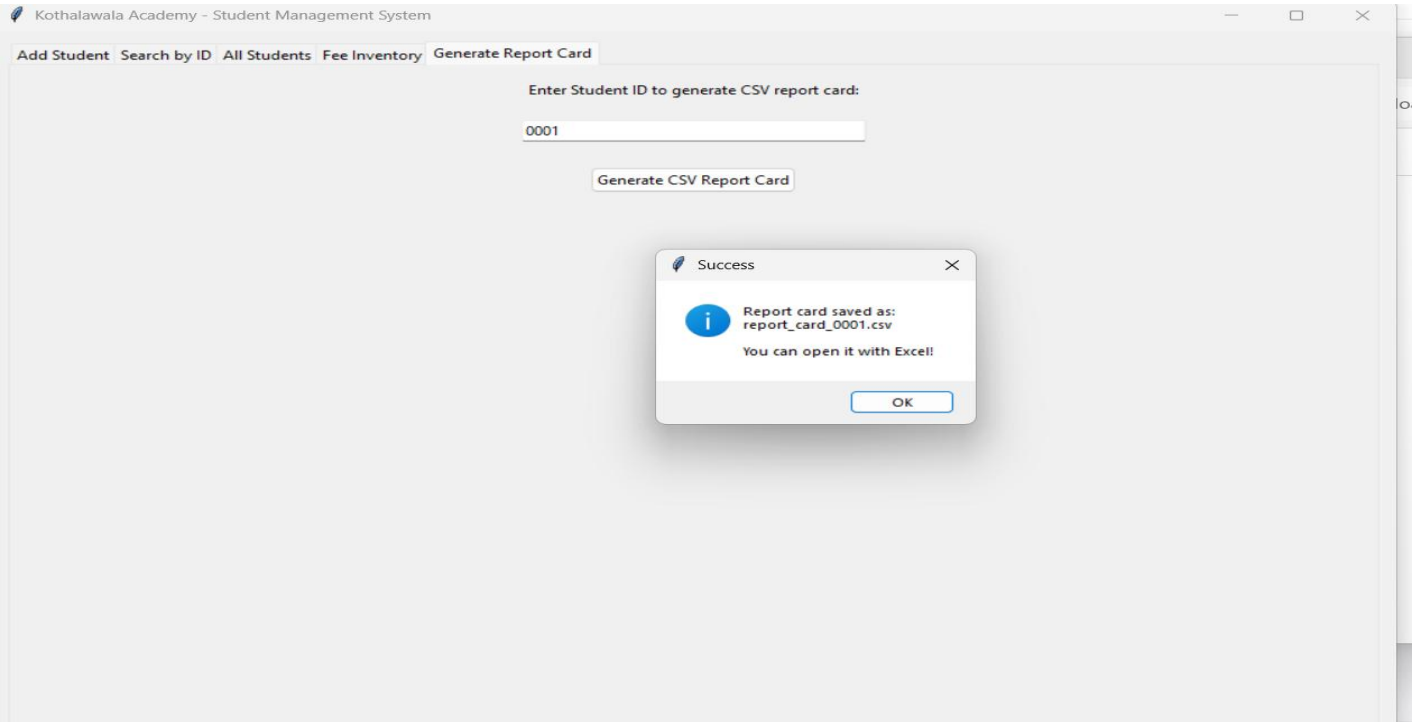
All Students List



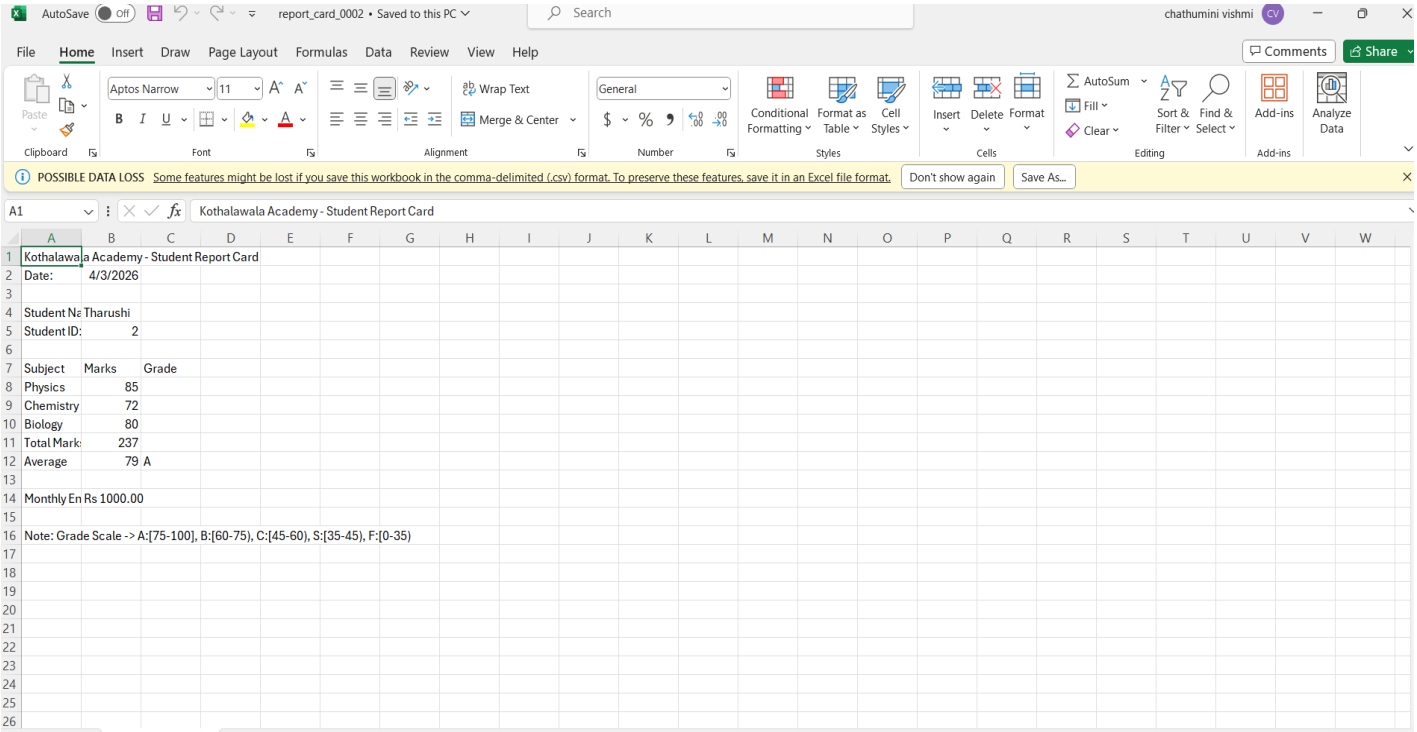
Fee Inventory Summary



Report Card opened in Excel.



Generated Report Card CSV



3.Technologies & Tools Used

Programming Language: Python 3

GUI Framework: Tkinter (built-in Python library for creating windows and tabs)

Data Storage: JSON (for saving and loading student records persistently)

Report Generation: CSV module (for creating downloadable report cards)

Development Environment: Visual Studio Code

How to Run the System Locally:

Install Python 3 on your computer (if not already installed).

Save the project file as SMS.py.

Double-click the file or open Command Prompt / Terminal and run:

python SMS.py

The GUI window will open immediately.

4.Team Contribution

TAR Gayanjalee - Designed GUI layout, main logic, and overall integration

WSCV Dananjalee - Implemented Student class, marks calculation & grade logic.

WGNR Priyadarshana - Developed fee inventory, total fee summary, and CSV report card feature

RMGGIN Aberathna - Added search functionality, input validation & error handling.

WMHR Ilangasinghe - Testing, debugging, documentation, and final report preparation.

5.Repository

GitHub Repository Link:

Complete codebase (SMS.py) has been uploaded.

Sample data file (students. Json) and example CSV report cards are included.

A detailed README.md file with setup and run instructions is provided.